UNIVERSITY OF WOLLONGONG AUSTRALIA

Tutorial 10 — Optional

P, PI and PID: design and implementation checks

Optional controller-design study after the eight-class core course, separate from Project 2. Suggested 60-minute tutorial: Q1 (6 min), Q2 including MATLAB (30 min), Q3 (14 min), and discussion and feedback (10 min). Extensions are additional practice. Start independently, explain one design choice to a partner, then attempt the PID audit independently.

Question 1 Controller actions and saturation

Give a brief reason or calculation for each part.

a) Identify controller actions. Classify $C_1(s)=2$, $C_2(s)=2+0.1/s$ and $C_3(s)=2+0.1/s+0.5s$. Which term accumulates error and which responds to its rate?

b) A proportional command. A temperature controller has $K_p=2$ W/°C, error +0.4°C and no other active controller term. Find its requested power. Does a nonzero constant error make this power grow with time?

c) Differentiate the signal. With $K_d=0.5$ W·s/°C, find ideal derivative contribution for constant error +0.4°C and for error increasing at 2°C/s. Why does a reference step cause difficulty when differentiating error?

d) A heater is limited to 0–5 W. Its controller requests 8 W while the temperature error is positive. What power is delivered? Should conditional integration allow the integral contribution to keep increasing?

Question 2 A PI temperature controller with a finite heater

Allow about 30 min, including the MATLAB comparison.

The chamber temperature-rise plant below has gain in °C/W. The heater can deliver 0–5 W. The reference is a 12°C rise from a zero initial state. First use P gain K_p = 0.5 W/°C. Then design PI control for target poles −0.04 and −0.06 s^{-1}. Assume unity negative feedback; distinguish linear predictions from the limited actuator.

$$G(s)=\frac{3}{45s+1},\ C_{PI}(s)=K_p+\frac{K_i}{s}$$

a) For P control, find closed-loop transfer function, final temperature rise, steady error and initial power demand. Is the target physically attainable in steady state?

b) For PI control, match the polynomial and find both gains and units. Write the actual closed-loop transfer function with its numerator.

c) Find the PI initial and steady power demands. Use the plant differential equation to predict the initial temperature slope with and without the 5 W limit. Is a feasible steady demand enough to avoid startup saturation?

MATLAB comparison. Open Code/MATLAB/Tutorial10_PIStarter.m in the course folder. Enter your gains from part (b). The supplied model simulates linear control and the 0–5 W heater with conditional integration; complete the plotting section.

Plot temperature rise and power in two panels of one figure. Label the two cases and the heater limit. Check the initial slopes against part (c). Submit the figure and your script. Base MATLAB is sufficient.

Question 3 Audit two proposed PID settings

Allow about 14 min. Audit the proposed gains without repeating the PI design.

A separate normalized second-order process and ideal parallel PID are given below. An additive disturbance D enters at the plant input after the controller. Use unity negative feedback. Desired characteristic polynomial is $(s + 8)(s^2 + 6s + 25)$. Candidate A has $(K_d, K_p, K_i) = (10, 57, 200)$; candidate B has (14, 73, 200).

$$G(s)=\frac{1}{s^2+4s+16},\ C(s)=K_d s+K_p+\frac{K_i}{s}$$

a) The characteristic polynomial is $s^3 + (4 + K_d)s^2 + (16 + K_p)s + K_i$. Expand the desired polynomial and select the correct candidate. Identify the mistake in the other candidate.

b) For the correct candidate, write the reference and disturbance transfer functions and identify the closed-loop poles. Does a step disturbance leave a steady output offset?

c) Explain why a denominator match does not justify a zero-free second-order overshoot formula. What is the practical problem with ideal differentiation of a reference step?

Extension A Filtered derivative action

Further practice. Parts (a) and (d) use first-order frequency and step responses; part (b) is an advanced polynomial check.

Keep the correct PID gains of Question 3, but replace the ideal derivative with the filtered branch below. A noise sinusoid of amplitude 0.003 normalized output units at 100 rad/s is applied to that branch. Compare T_f = 0.05 s and 0.10 s. This branch calculation is not the total closed-loop noise response.

$$C_D(s) = \frac{K_d s}{1 + T_f s}$$

a) Calculate derivative-branch command amplitude for both filters and for an ideal derivative. Give the high-frequency gain limit.

b) Advanced check. Derive the filtered-PID characteristic polynomial. Use polynomial roots to check both T_f values. Does the original pole target remain unchanged?

c) Which closed-loop responses and implementation limits must be checked again after adding the filter?

d) For a unit step applied directly to the derivative input, find the filtered branch command and its initial peak for both filters. How does differentiating measured output avoid reference kick?

Extension B Exact linear PI response

For the unsaturated controller in Question 2, obtain the 12°C step response by partial fractions. Check its initial value, initial slope and final value. Does this expression also describe the power-limited startup?

Method reference

Concept references: Ogata, System Dynamics, 4th ed., Sec. 10–3, printed pp. 501–505; Modern Control Engineering, 5th ed., Sec. 8–5, printed pp. 590–592. The conditional-integration example uses the stated heater limits.

Method source: Ogata, Modern Control Engineering, 5th ed., Problem A-8-5, pp. 617–619 (PID design). The thermal PI design, candidate audit and conditional-integration tests below are independently adapted. Page numbers refer to printed pages.